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METRIC SPACE

Real analysis → $\mathbb{R}$ → func
- limit
- cont
- dist we will find

Algebra → S → dist → metric space.

Metric Space :-

Triangle Inequality

$|\alpha-\beta| \leq |\alpha-\gamma|+|\gamma-\beta|$

A non-empty set X with a map $d : X \times X \to \mathbb{R}$ is called metric space if the map d has the following properties

i) $d(x,y) \geq 0 \quad \forall\ x,y \in X$

ii) $d(x,y) = 0 \iff x = y$

iii) $d(x,y) = d(y,x) \quad \forall\ x,y \in X$

iv) $d(x,y) \leq d(x,z) + d(z,y) \quad \forall\ x,y,z \in X$

→ Triangle inequality

$d \to$ metric on X

$(X,d) \to$ metric space

Ex :- Let $X = \mathbb{R}$

$d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}^+$ defined by

$$d(x,y) = |x-y|$$

Sol

i) $|x-y| \geq 0 \Rightarrow d(x,y) \geq 0$

ii) $d(x,y) = 0 \Leftrightarrow |x-y| = 0$

$\Leftrightarrow x = y$

iii) $d(x,y) = |x-y| = |y-x|$

$= d(y,x)$

iv) $d(x,y) = |x-y|$

$= |x-z+z-y|$ $(\because |a+b| \leq |a|+|b|)$

$\leq |x-z| + |z-y|$

$= d(x,z) + d(z,y)$

$\therefore d(x,y) \leq d(x,z) + d(z,y)$

$\boxed{\therefore\ d(x,y) = |x-y|}$ is a metric space

$\hookrightarrow$ usual / standard metric space.

Example :-

Let $X = \mathbb{R}^2$ and d is defined as

$d : X \times X \to \mathbb{R}$

$$d((x_1, x_2), (y_1, y_2)) = \underbrace{\left[(x_1-y_1)^2 + (x_2-y_2)^2\right]^{1/2}}$$

§ $x = (x_1, x_2)$ and $y = (y_1, y_2)$ — dist fun. in $\mathbb{R}^2$

$z = (z_1, z_2)$

1) To show $d(x,y) \geqslant 0$

$$d(x,y) = \Big[\underbrace{(x_1-y_1)^2}_{\geqslant 0} + \underbrace{(x_2-y_2)^2}_{\geqslant 0}\Big]^{1/2}$$

$\geqslant 0$

2) To show $d(x,y) = 0 \iff x = y$

$d(x,y) = 0$

$\iff \left[(x_1-y_1)^2 + (x_2-y_2)^2\right]^{1/2} = 0$

sq on b.s

$\iff (x_1-y_1)^2 + (x_2-y_2)^2 = 0$

sum of two +ve number is 0 only when both are 0

$\iff x_1 - y_1 = 0$ and $x_2 - y_2 = 0$

$\iff x_1 = y_1$ and $x_2 = y_2$

$\iff x = y$

3) To show $d(x,y) = d(y,x)$

$$d(x,y) = \left[(x_1-y_1)^2 + (x_2-y_2)^2\right]^{1/2}$$

$$= \left[(y_1-x_1)^2 + (y_2-x_2)^2\right]^{1/2}$$

$$= d(y,x)$$

4) $d(x,y) \le d(x,z) + d(z,y)$

Let $a_1 = x_1 - z_1$, $a_2 = x_2 - z_2$

$b_1 = z_1 - y_1$, $b_2 = z_2 - y_2$

$$d(x,y) = \left[(a_1+b_1)^2 + (a_2+b_2)^2\right]^{1/2}$$

$$= \left[\sum_{k=1}^{2} (a_k+b_k)^2\right]^{1/2}$$

$$d(x,z) = \left(a_1^2 + a_2^2\right)^{1/2} = \left(\sum_{k=1}^{2} a_k^2\right)^{1/2}$$

$$d(z,y) = \left(b_1^2 + b_2^2\right)^{1/2} = \left(\sum_{k=1}^{2} b_k^2\right)^{1/2}$$

To show

$$\left[\sum_{k=1}^{2} (a_k+b_k)^2\right]^{1/2} \le \left(\sum_{k=1}^{2} a_k^2\right)^{1/2} \left(\sum_{k=1}^{2} b_k^2\right)^{1/2}$$

sq on b.s

$$\sum_{k=1}^{2} (a_k+b_k)^2 = \left[\left(\sum_{k=1}^{2} a_k^2\right)^{1/2} + \left(\sum_{k=1}^{2} b_k^2\right)^{1/2}\right]^2$$

Sq on b.s

$$\sum_{k=1}^{2} \left(\cancel{a_k^2} + \cancel{b_k^2} + \cancel{2}a_k b_k\right) = \sum_{k=1}^{2} \cancel{a_k^2} + \sum_{k=1}^{2} \cancel{b_k^2} \cancel{+2}\left(\sum_{k=1}^{2} a_k^2 b_k^2\right)^{1/2}$$

$$\sum_{k=1} a_k b_k \le \left(\sum_{k=1} a_k^2\right)^{1/2} \left(\sum_{k=1} b_k^2\right)^{1/2}$$

↳ which is Cauchy Schwarz inequality

⇒ True for every real no.

Given d is

∴ Metric space.

Ex 2: Let $X = \mathbb{R}^n$

$x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$

$y = (y_1, y_2, \ldots, y_n) \in \mathbb{R}^n$

$$d(x,y) = \left[\sum_{i=1}^{n} (x_i - y_i)^2\right]^{1/2}$$

$$d_\infty(x,y) = \max_{1 \le i \le n} |x_i - y_i|$$

Sol

i) $|x_i - y_i| \geq 0$

$\Rightarrow \max_{1 \le i \le n} |x_i - y_i| \geq 0$

$\Rightarrow d_\infty(x,y) \geq 0$

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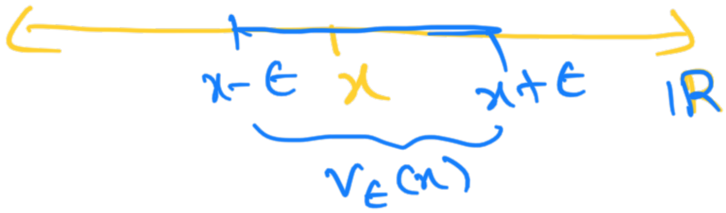
Classes of open and closed set :-

Neighbourhood of a point

A nbd of a point $x \in \mathbb{R}$ is any set V that contains an ϵ-nbd

$$V_\epsilon(x) = \{ y \,/\, |y - x| < \epsilon \} = (x-\epsilon, x+\epsilon)$$

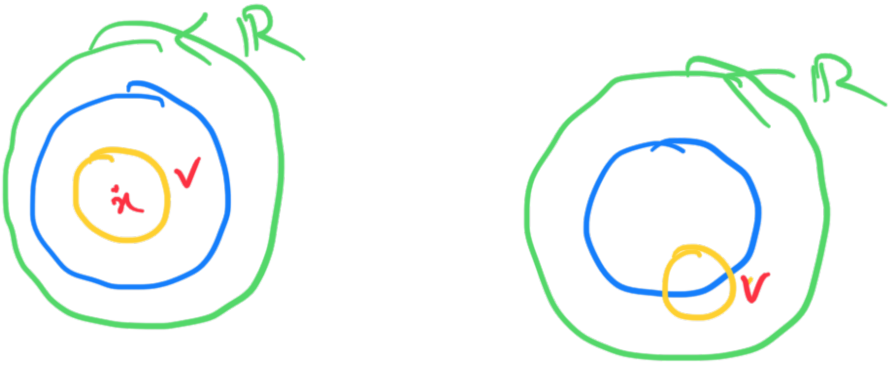
of x for some $\epsilon > 0$



Open set

A subset G of $\mathbb{R}$ is open in $\mathbb{R}$ if for each $x \in G$ $\exists$ a nbd

V of x s.t V ⊆ G



NOTE:-

To show $G \subseteq \mathbb{R}$ is open, it suffices to show that each point in G has an ϵ-nbd contained in G

G is open if $x \in G$ for some $\epsilon > 0$ $(x-\epsilon, x+\epsilon) \subseteq G$

Ex 1: The entire $\mathbb{R}$ is open set in $\mathbb{R}$

Basically $\mathbb{R} \subseteq \mathbb{R}$

where $\mathbb{R} = (-\infty, \infty)$

We have to show $\mathbb{R}$ is open

As we know $\mathbb{R}$ is in open interval

But we have to show $\mathbb{R}$ is open set

i.e we have to take any element from this set and we have to show

Let $x \in \mathbb{R}$ be an arbitrary element

Take $\epsilon_x = 1$

$(x-\epsilon_x, x+\epsilon_x) = (x-1, x+1) \subseteq \mathbb{R}$

$(-\infty, \infty)$ in $\mathbb{R}$
i.e all elements contain in $\mathbb{R}$

in this interval for every element has ϵ-nbd exists
i.e all contain in $\mathbb{R}$
so let us see for one element.

$\therefore$ $\mathbb{R}$ is open

$\epsilon = 1$

$\epsilon = 2, 3, 1/2$ any can take
but condition need to check
$\epsilon > 0$ real number take
i.e $\epsilon \geq 0$
and check.

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Ex 2 The set $G = \{x \in \mathbb{R} \mid 0 < x < 1\}$ is an open set in $\mathbb{R}$.

Soln $G = (0, 1)$

Let $x \in G$ be an arbitrary element

x is near to x

we have to show ϵ-nbd is near x is there which is containing

case 1

Let $x = 1/2$

$x = 1/2$ Defined 2 means → $\epsilon = 1/4$ can take

($1/2$ is less than so if $1/2$ more means it goes more than 0

Becoz ϵ-nbd should be b/w 0 to 1)

Take $\epsilon = 1/4$

$$(x-\epsilon_x, x+\epsilon_x) = \left(\tfrac{1}{2}-\tfrac{1}{4}, \tfrac{1}{2}+\tfrac{1}{4}\right)$$

$$= \left(\underset{>0}{\tfrac{1}{4}}, \underset{<1}{\tfrac{3}{4}}\right)$$

$$\subseteq (0,1)$$

$$= G$$

This this ϵ-nbd contained in G

$\Rightarrow$ G is open

Case (ii):- $x \neq \frac{1}{2}$

no fixed x can't think so ϵ_x can be any.

Take $\epsilon_x = \min\{x, 1-x\}$ 0 to 1 dist = 1

we took x is near 0

To show:-

$(x-\epsilon_x, x+\epsilon_x) \subseteq (0,1) = G$

So $\epsilon_x = \min$ & $1/2 > 0$ $\epsilon_x = 1/2$ means.

i.e min of $(x, 1-x)$

Let $y \in (x-\epsilon_x, x+\epsilon_x)$

be any element

$$x-\epsilon_x < y < x+\epsilon_x$$

(i.e if $0 \le x-\epsilon_x < y < x+\epsilon_x \le 1$) [To show: $y \in (0,1)$]

we are done $\square$ (1)

Case I :- Let $x = \min\{x, 1-x\} = \epsilon_x$

$x < 1-x$

$x - \epsilon_x = x - x = 0$

$x + \epsilon_x = x + x < \cancel{x} + 1 - \cancel{x} = 1$

$x + \epsilon_x < 1$

from (1)

$x - \epsilon_x < y < x + \epsilon_x < 1$

$0 < y < 1 \Rightarrow G$ is open

Case II :- Let $1-x = \min\{x, 1-x\} = \epsilon_x$

$x - \epsilon_x = x - (1-x) = x - 1 + x$

$1 - x < x$ $\qquad = 2x - 1 > 0$

$1 < 2x$

$0 < 2x - 1$ or $2x - 1 > 0$

$x + \epsilon_x = x + (1-x) = 1$

from (1)

$0 < x - \epsilon_x < y < x + \epsilon_x = 1$

$0 < y < 1$

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$\Rightarrow G$ is open.

$\because$ for $x = 1/2$ and $x \neq \frac{1}{2}$

G is open.

$\therefore$ G is open set in $\mathbb{R}$.

H/W

Any open interval $I = (a, b)$ is an open set in $\mathbb{R}$.

Sol. $x \in (a, b)$

$(x - \epsilon_x, x + \epsilon_x) \in (a, b)$

for finding ϵ_x

$\epsilon_x = \min\{x - a, b - x\}$

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** All the best **